

Complete project structure

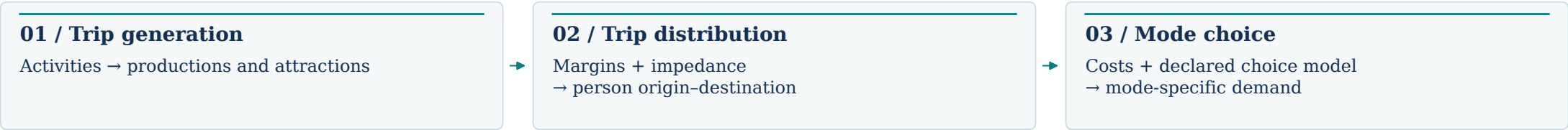
SOURCE EVIDENCE

Roads, boundaries, zones Network geography records	People and activity Households, population, proxies	Transit and walking GTFS, fares, service and access	Observation records GPS, trajectories, detectors
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MODEL REPRESENTATION AND UPSTREAM PREPARATION

GMNS representation Directed physical links Hierarchy, centroids, access Source IDs, units and turns	City demand preparation Statistics and boundaries Separate activity attributes Version / geography checks	Declared vehicle OD Supplied demand may enter static assignment directly No demographic compiler	Observation association Match/associate links and service Optional input or diagnostics Not automatic OD or calibration
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FOUR-STAGE TRAVEL-DEMAND WORKFLOW



04 / Traffic assignment
Vehicle demand → network flows

FOUR INTERNAL ASSIGNMENT LAYERS

A / Native assignment Reference solvers and reconstruction	B / Decomposition Distributed methods and subproblems	C / Spatial hierarchy Representation and access / turn states	D / Coordination Shared capacities and verification
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TWO INDEPENDENT MATHEMATICAL CONTRACTS

Static road assignment BPR / Beckmann; period-specific vehicle OD Frank-Wolfe · official tap-b Algorithm B; adapters vary Finite-path reference · native L3 diagnostics Physical-link flows and original-space checks	Finite time-expanded optimization Separately selected graph, OD, time and capacities Fixed arc costs · shared hard capacity Arc-flow LP · two-phase CG · Lagrangian · ADMM Movement-flow reconstruction and feasibility checks
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SAVED OUTPUTS AND INDEPENDENT CHECKS

Saved outputs and independent checks Zonal / mode demand · generated columns · physical-link flows · CSV / SQLite · figures Units, conservation, objective / gap and provenance checks are specific to each instance.
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REAL-CITY CASES / ONE COMPARISON FRAMEWORK

Boston City demand and static scales Separate bounded finite cases	Hong Kong Turn-aware city demand Separate static and finite cases	Ann Arbor 25-zone morning city model S600 static; finite unsolved	Urbana-Champaign 21-zone morning city model S72 static; separate T4
Ithaca 11-zone midday activity model S110 static; separate T4	Berkeley 13-zone morning city model S72 static; separate T4	Chicago 5-zone morning city model S20 static; separate T4	Pittsburgh 41-group morning city model S72 static; separate T4

DEMONSTRATION BENCHMARK

Sioux Falls Supplied-OD static benchmark · separate historical selected 200 / 250-OD finite cases City-demographic and mode-choice preparation are outside this benchmark.

TOOLS AND DOCUMENTATION

Data and GMNS tools Catalog · exports · queries	Generic RC5 engine 0.3.0-rc5 scope only	Versioned method code Algorithms · case adapters	Examples and checks Saved results · validation Independent validators
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